Introduction to Computer Networks

Dr Vijay Ukani

Chapter 1 Overview

Based on Computer Networks, Fifth Edition
by Andrew Tanenbaum and David Wetherall
Pearson Education-Prentice Hall, 2011

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Uses of Computer Networks

Business Applications

- Company LANs
- Resource Sharing
- Virtual Private Networks (VPNs)
- Web Applications & e-commerce
- Client-server model
- Email & VoIP

Home Applications

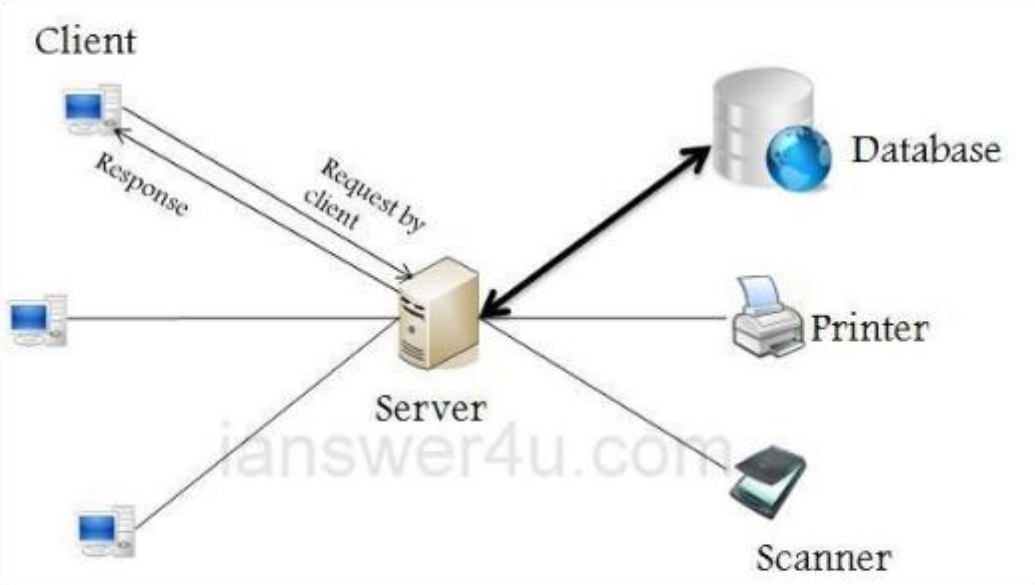
- Internet Access
- Client-server and peer-to-peer (P2P)
- Messaging & Social Media
- IPTV
- Smart home technologies
- Forms of e-commerce

Business Applications: Client-Server Model

Business Applications (1): A network typically consists of multiple clients communicating with one or more servers.

Business Applications (2): The client-server model fundamentally involves *requests* sent by clients and *replies* provided by servers.

This architecture is the foundation for web applications, email, and corporate databases.

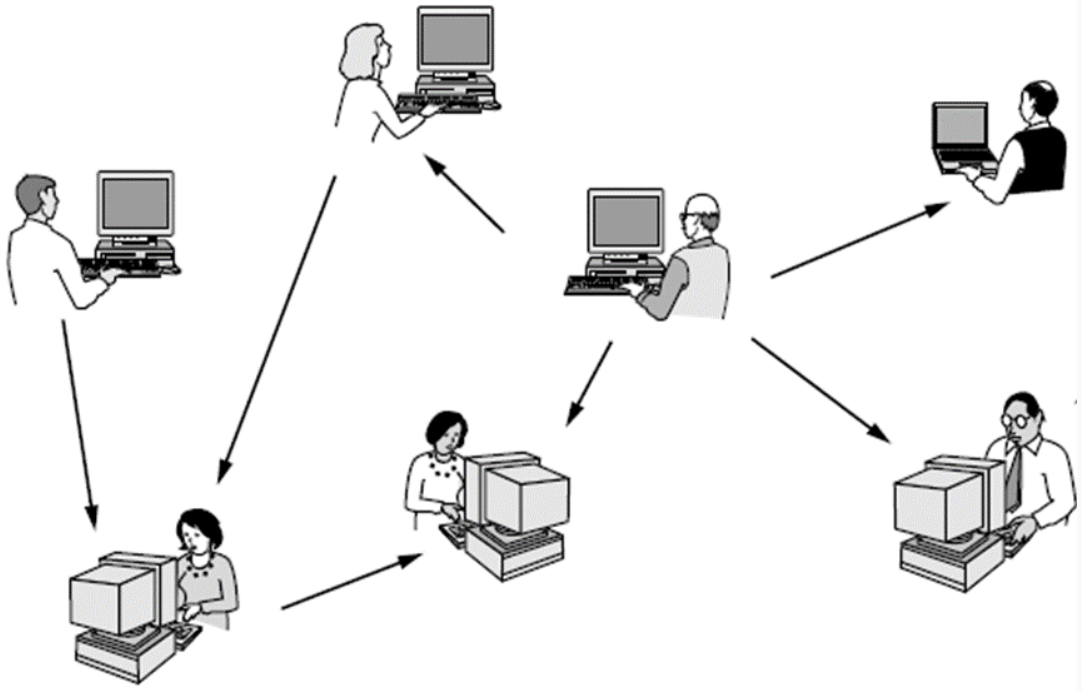


ianswer4u.com

Home Applications: Peer-to-Peer & E-commerce

Home Applications (1): In a peer-to-peer (P2P) system, there are no fixed clients and servers. Every node can act as both a client and a server simultaneously.

Home Applications (2): Networks enable various forms of e-commerce, directly connecting consumers to digital marketplaces from their homes.



Home Applications: Forms of E-commerce

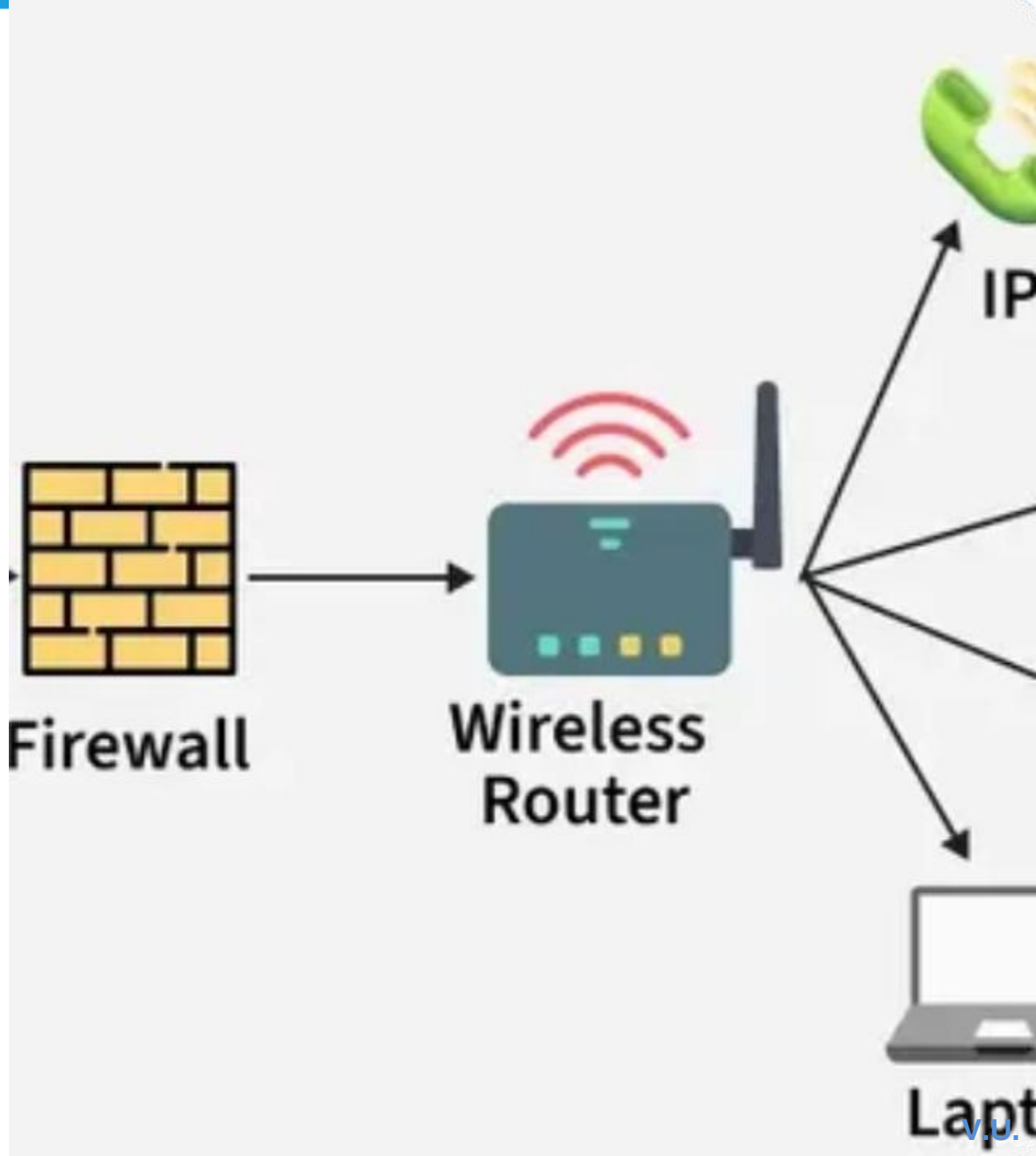
Tag	Full name	Example
B2C	Business-to-consumer	Ordering books online
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products online
P2P	Peer-to-peer	Music sharing

Mobile Users

Wireless & Mobile Computing

Mobile users rely on combinations of wireless networks and mobile computing to stay connected globally.

- Constant Connectivity & Hotspots
- Cellular Networks & SMS
- GPS Navigation
- M-commerce (Mobile Commerce)
- RFID, NFC, and Sensor networks
- Wearable technologies



Mobile Users

Combination of Wireless & Mobile Computing

Wireless	Mobile	Typical applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in unwired buildings
Yes	Yes	Store inventory with a handheld computer

"Network neutrality means all Internet traffic should be treated equally independent of the source, without favoring or blocking particular product or site."

— Social Issues and Applications

Digital Millennium Copyright Act

User Privacy

Phishing

— Social Issues and Applications

Network Hardware

Exploring classification by scale, transmission technologies, and delivery models.

Network Hardware: Transmission & Delivery

Transmission Technology

Networks can be broadly categorized by their transmission technology:

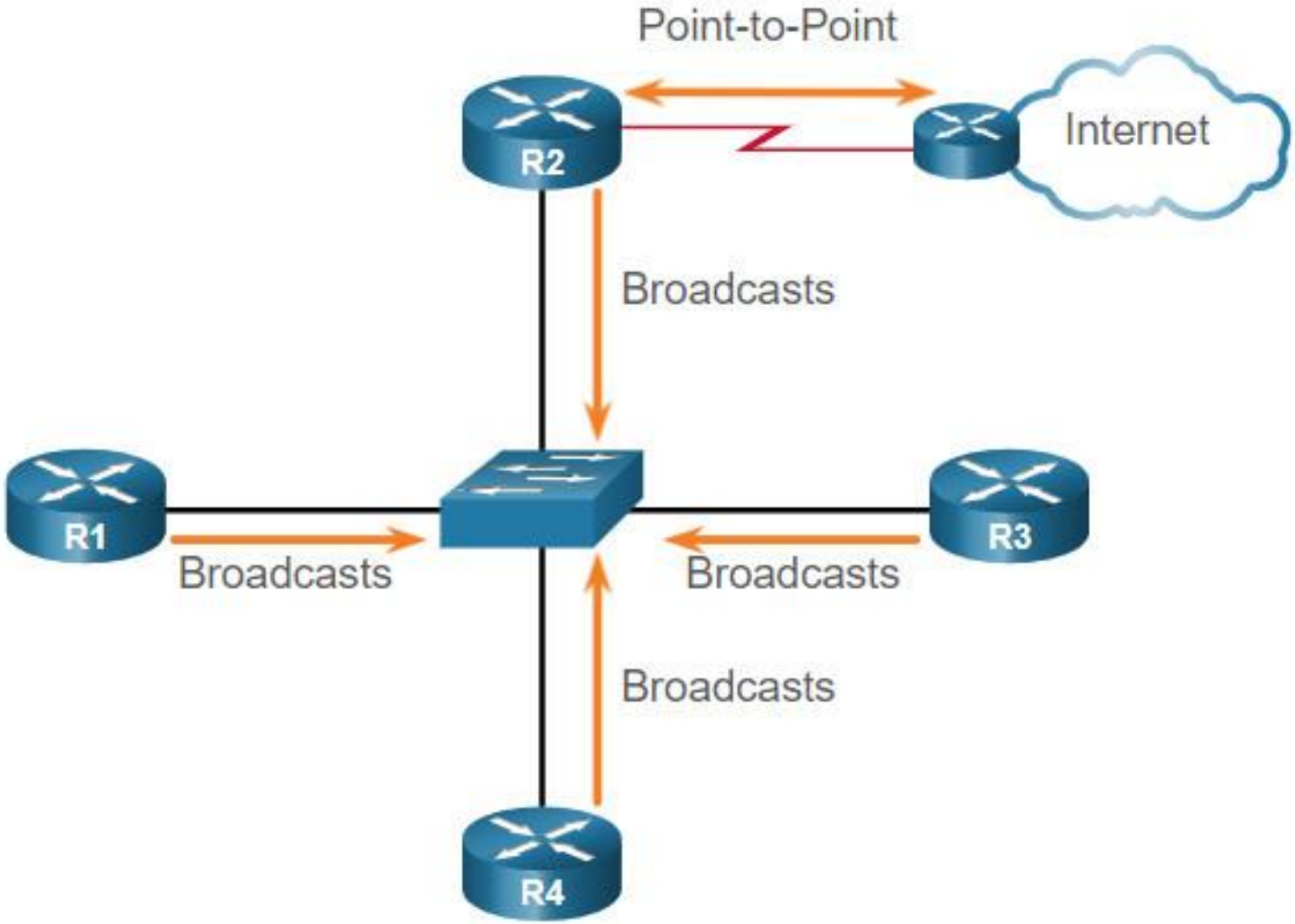
- Broadcast links: Communication channel shared by all machines on the network.
- Point-to-point links: Connections between individual pairs of machines.

Delivery Models

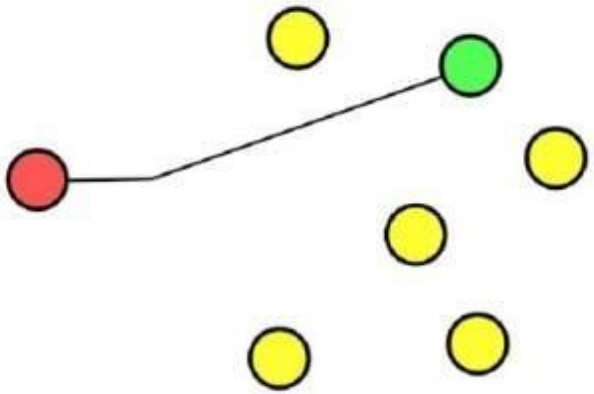
How data is addressed and delivered:

- Unicast: One-to-one delivery.
- Multicast: One-to-many (specific group).
- Broadcast: One-to-all.
- Anycast: One-to-nearest (of a specific group).

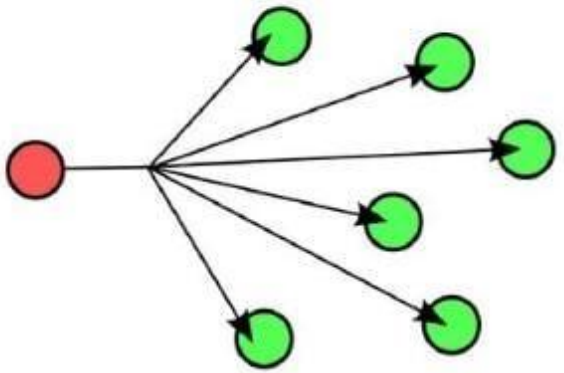
Network Hardware: Transmission Technology



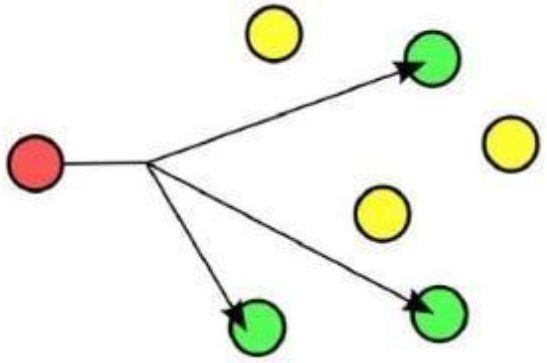
Network Hardware: Delivery Models



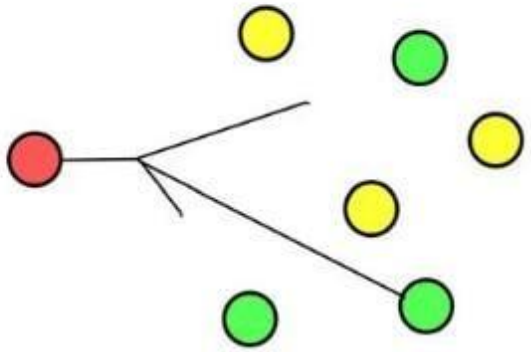
Unicast



Broadcast



Multicast



Anycast

Network Hardware: Classification by Scale

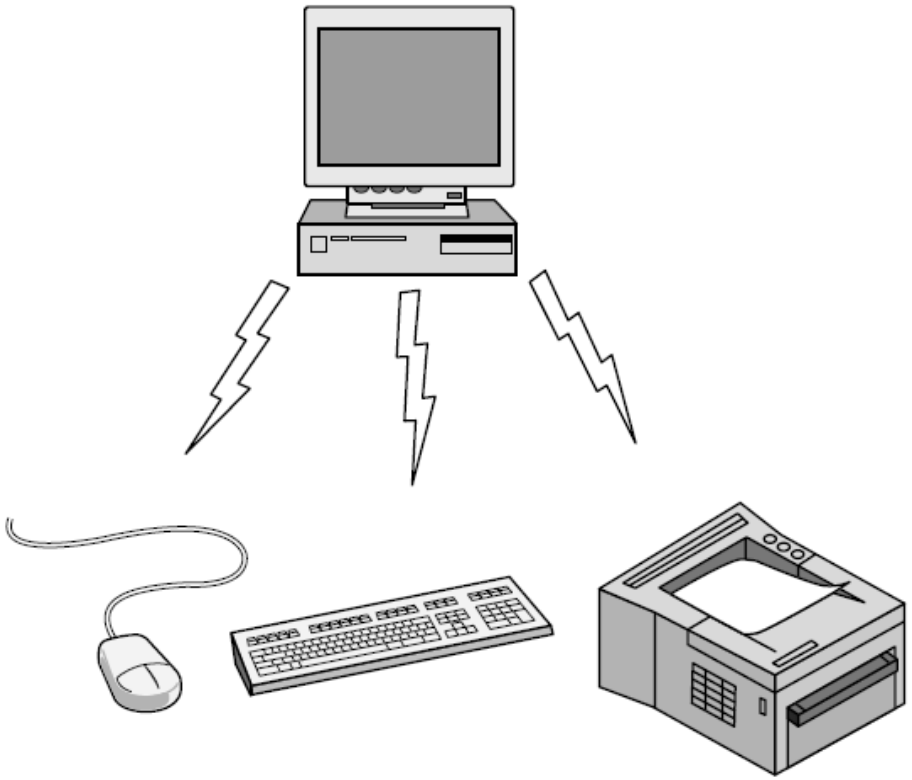
Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Network Hardware: Classification by Scale

Interconnected processors are classified primarily by their geographical scale and scope:

- Personal Area Networks (PAN): Within reach of a person (e.g., Bluetooth).
- Local Area Networks (LAN): A room, building, or campus.
- Metropolitan Area Networks (MAN): A city or large suburban area.
- Wide Area Networks (WAN): A country or continent.
- The Internet: A network of networks spanning the planet.

Personal Area Network (PAN)



Bluetooth PAN Configuration

A Personal Area Network (PAN) lets devices communicate over the range of a person.

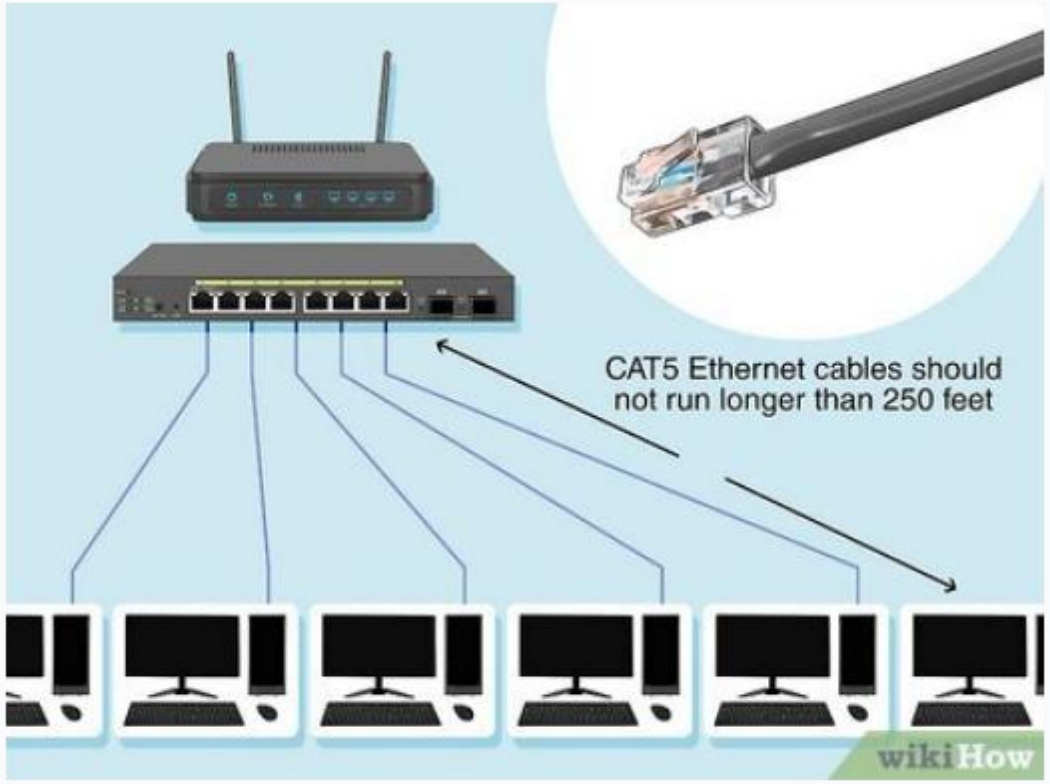
A common example is a wireless network that connects a computer with its peripherals. Bluetooth is the dominant technology for wireless PANs, creating a short-range, ad hoc network between master and slave devices.

Local Area Networks (LAN)

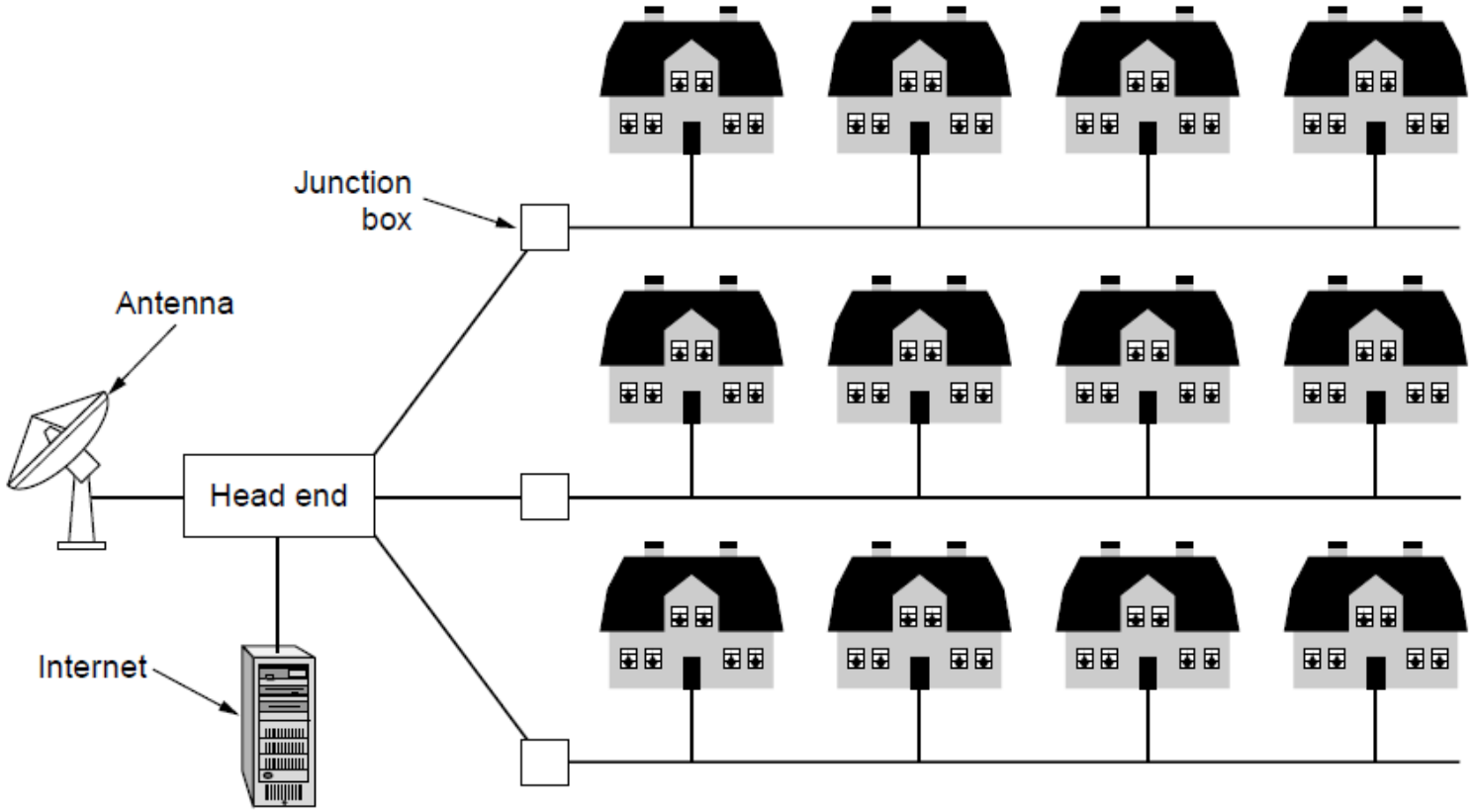
LANs are privately owned networks within a single building or campus of up to a few kilometers in size.

They are widely used to connect personal computers and workstations to share resources.

- (a) Wireless LANs: Typically use the 802.11 standard (WiFi) for connectivity without cables.
- (b) Wired LANs: Rely heavily on Switched Ethernet, using physical cables connected to a central switch.



Metropolitan Area Networks (MAN)



Based on Cable TV

A Metropolitan Area Network covers a city. The best-known example of a MAN is the cable television network available in many cities.

These systems grew from early community antenna systems into two-way, high-speed broadband networks providing Internet access to metropolitan areas.

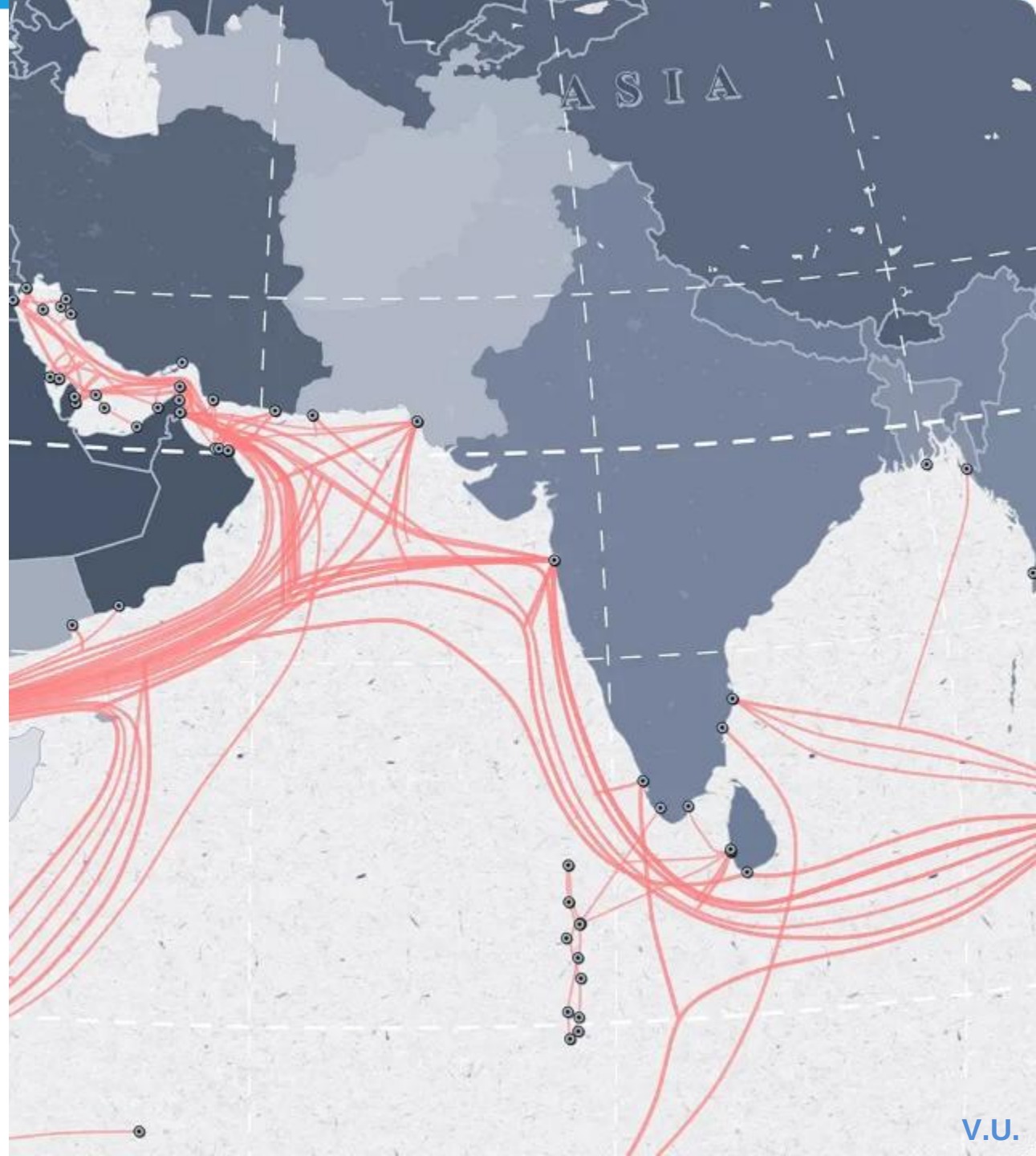
Wide Area Networks (WAN)

Connecting the Globe

WAN (1): **A WAN spans a large geographical area**. For example, a WAN that connects three branch offices in Australia.

WAN (2): Companies often create a WAN using a Virtual Private Network (VPN) over the public Internet for secure, cost-effective routing.

WAN (3): Alternatively, a WAN can be constructed using an ISP network, leveraging telecommunications infrastructure.



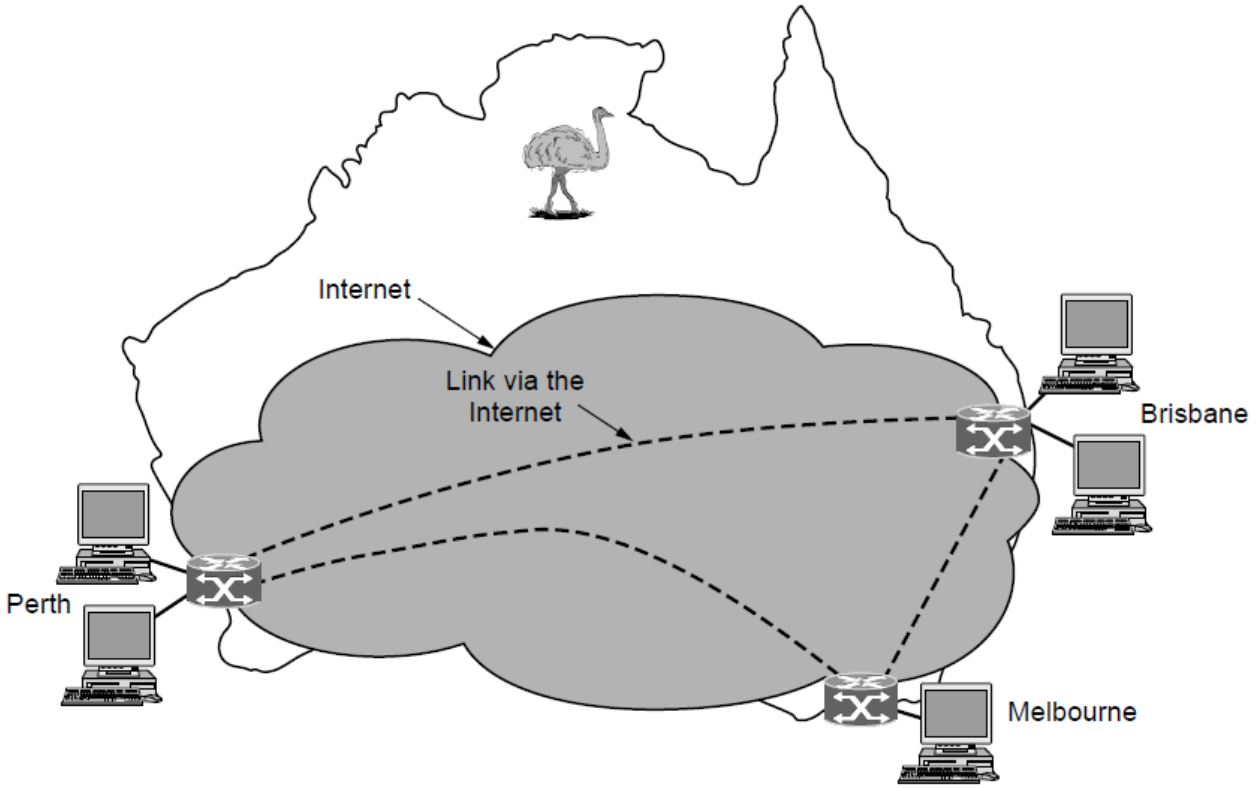
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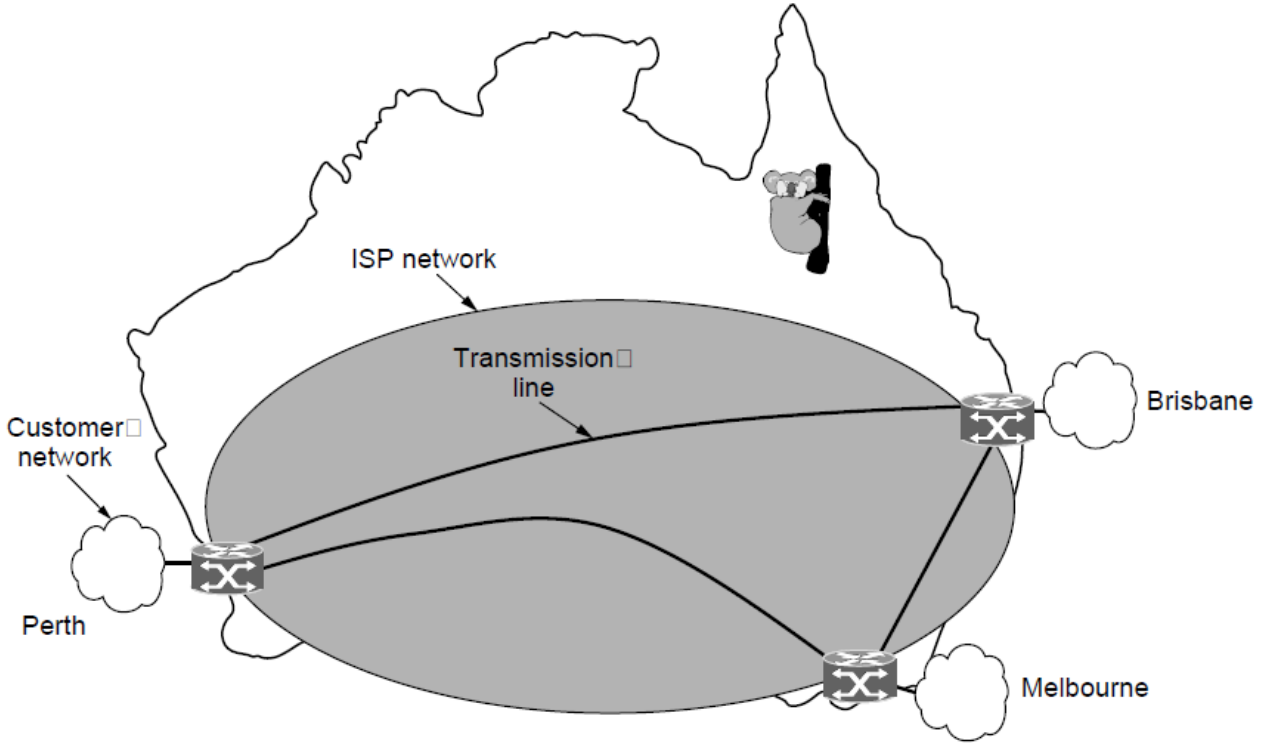
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Network Software

Protocol hierarchies, design issues, service primitives, and connections.

Key Terms



Packet & Datagram

A Packet is a formatted unit of data. In connectionless networks, it's specifically called a Datagram.



Routing & Switching

Packet switching forwards packets individually. Routing determines the path those packets take.



Virtual Circuit

A connection-oriented path established before any data is sent, mimicking a physical circuit.

Protocol Hierarchies

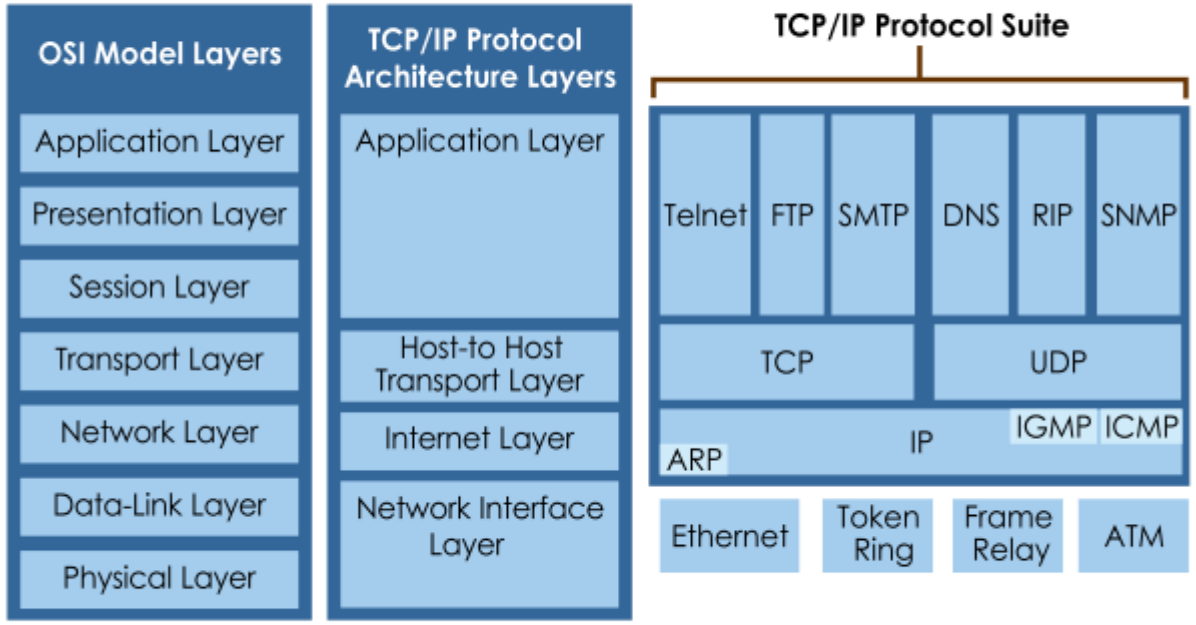
Protocol Hierarchies (1): Networks are organized as a stack of layers or levels, utilizing protocols and interfaces to separate concerns.

Protocol Hierarchies (2): The philosopher-translator-secretary architecture illustrates how peers communicate logically while passing data physically down the stack.

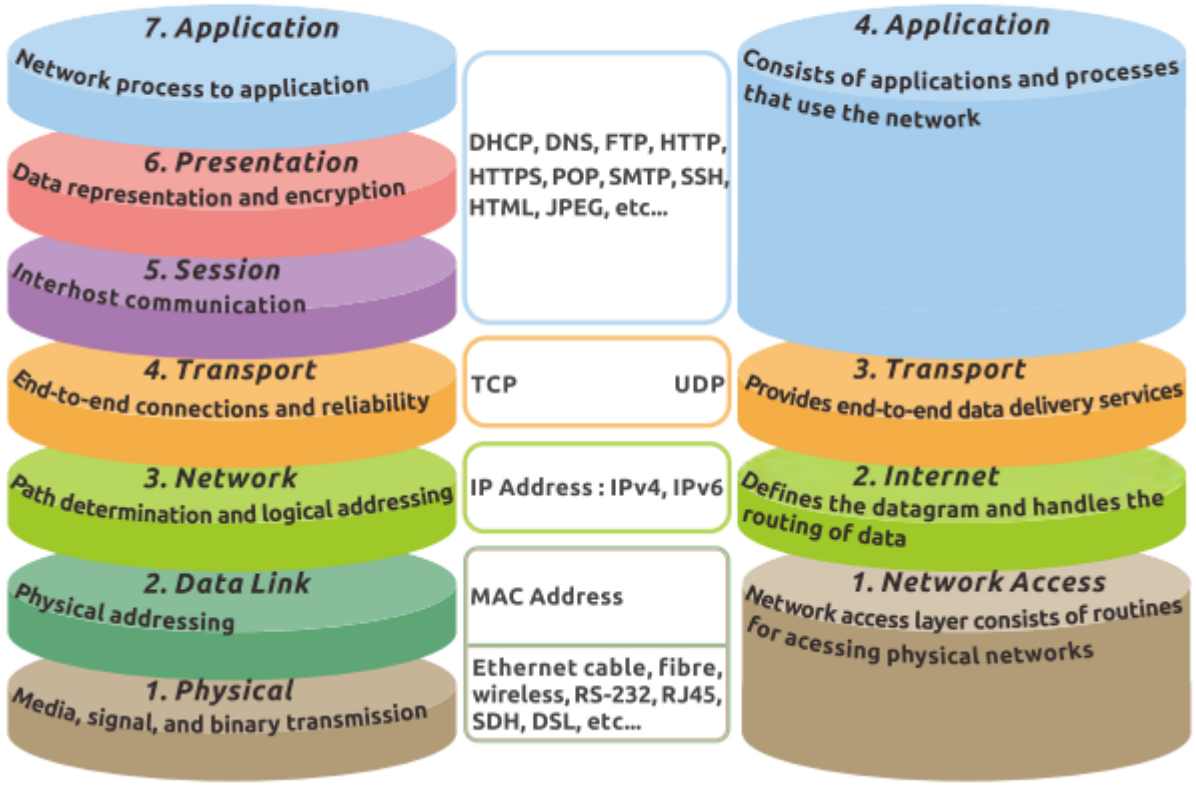
Protocol Hierarchies (3): Example information flow supporting virtual communication in layer 5 requires actual transmission through layer 1.

Protocol Hierarchies

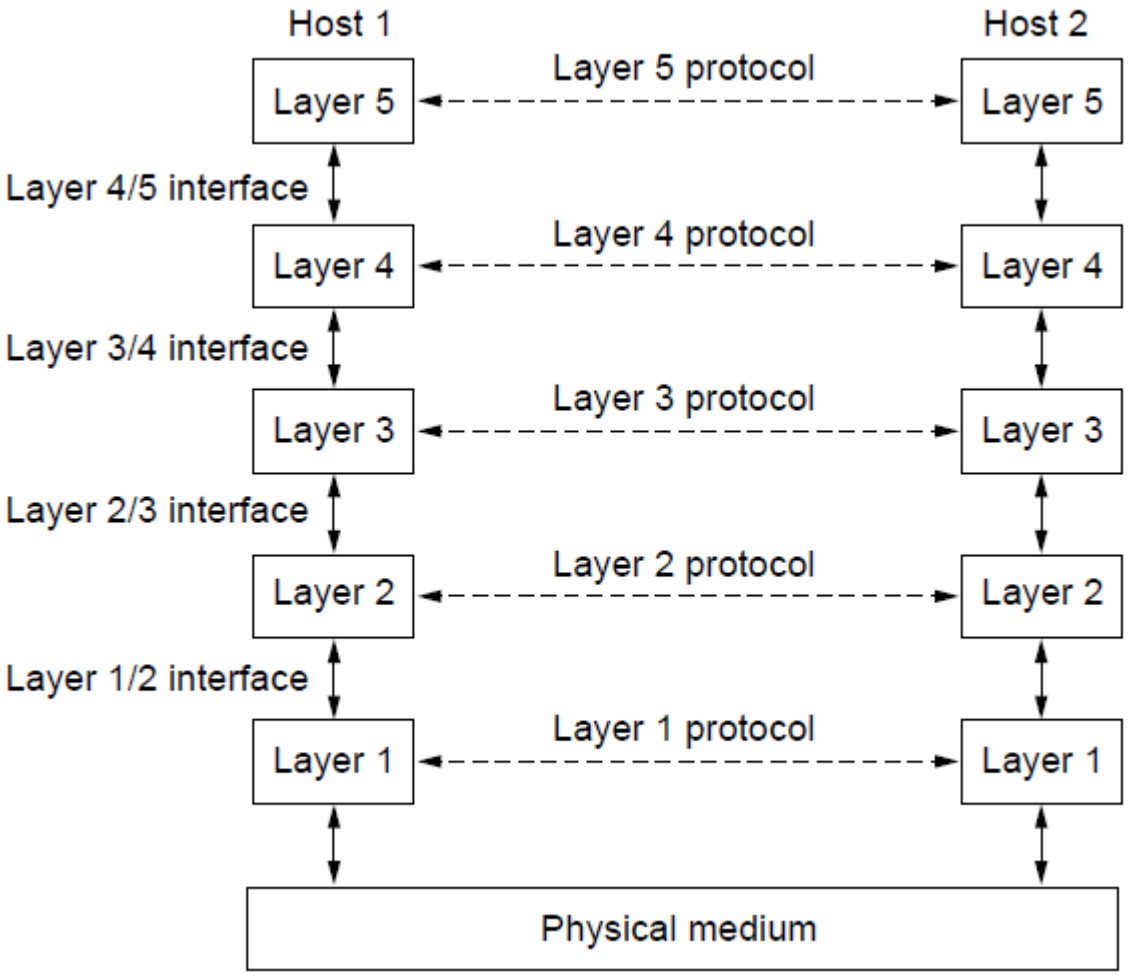
TCP/IP Protocol Architecture



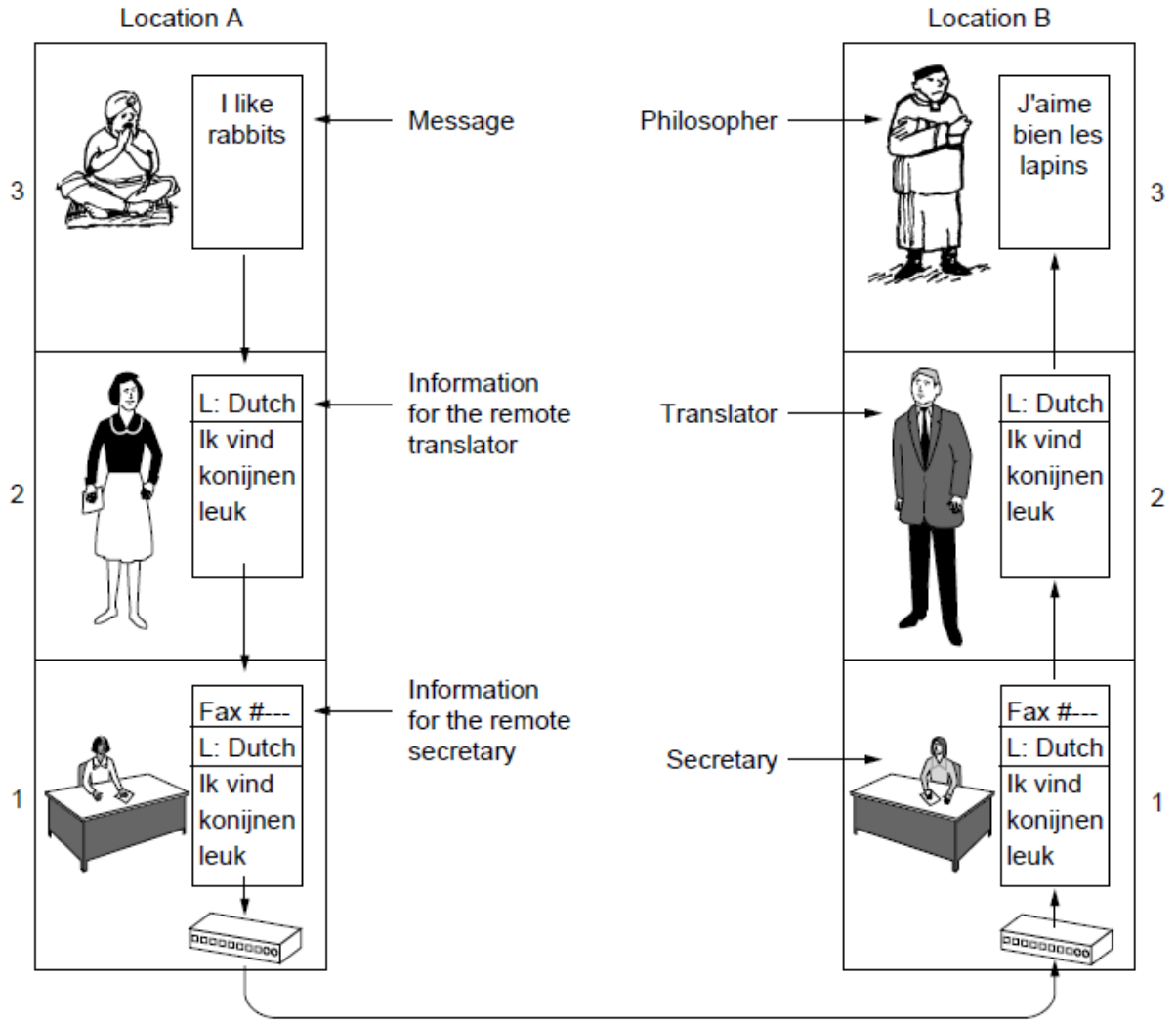
OSI Model vs. TCP/IP Model



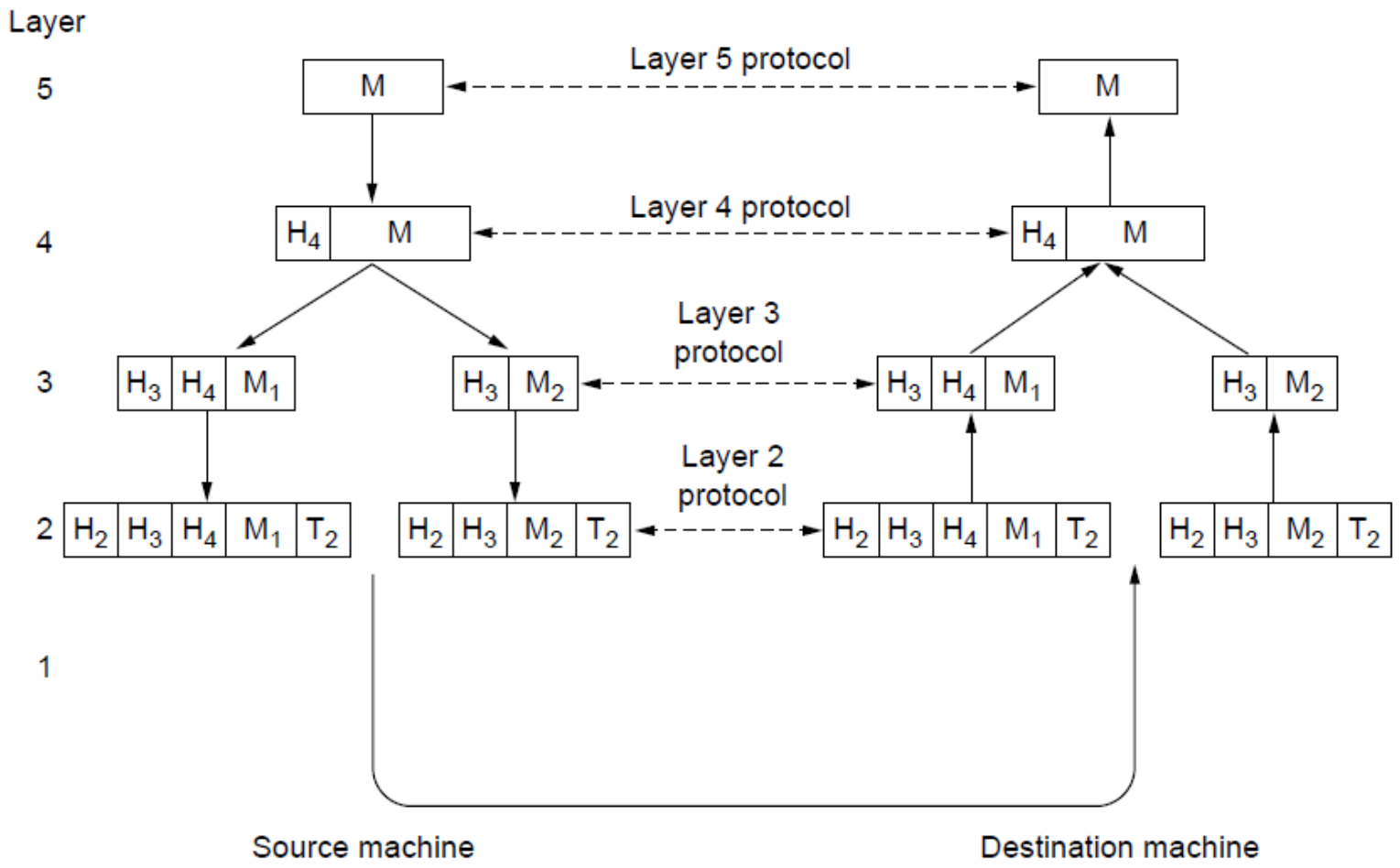
Protocol Hierarchies: Layers, Protocols and Interfaces



Protocol Hierarchies: Philosopher-translator-secretary architecture



Protocol Hierarchies: Virtual Communication between Peer Layers



Service Types: Connection-Oriented vs Connectionless

Connection-Oriented Service

Modeled after the telephone system. The user establishes a connection, uses it, and then releases it.

Includes types like reliable message stream, reliable byte stream, and unreliable connection.

Connectionless Service

Modeled after the postal system. Each message carries the full destination address and is routed independently.

Includes types like unreliable datagram, acknowledged datagram, and request-reply.

Service Types: Connection-Oriented vs Connectionless

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Service Primitives & Protocols

Service Primitives (1): Six service primitives that provide a simple connection-oriented service:


LISTEN: Block waiting for an incoming connection.


CONNECT: Establish a connection with a waiting peer.


RECEIVE: Block waiting for an incoming message.


SEND: Send a message to the peer.

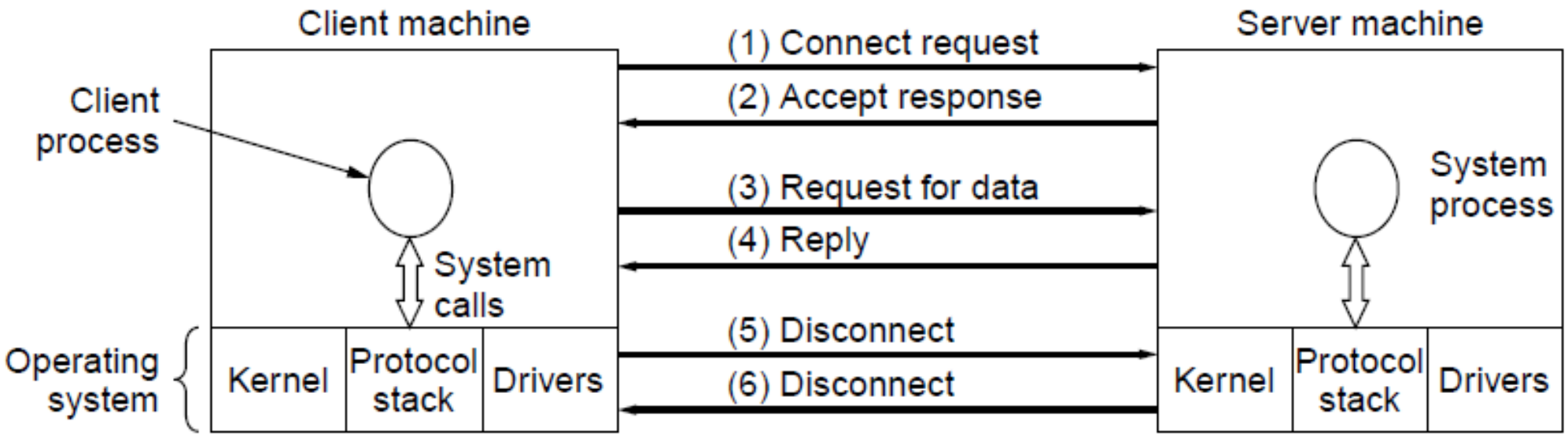

DISCONNECT: Terminate a connection.

Service Primitives (2): A simple client-server interaction can use acknowledged datagrams instead of full connections.

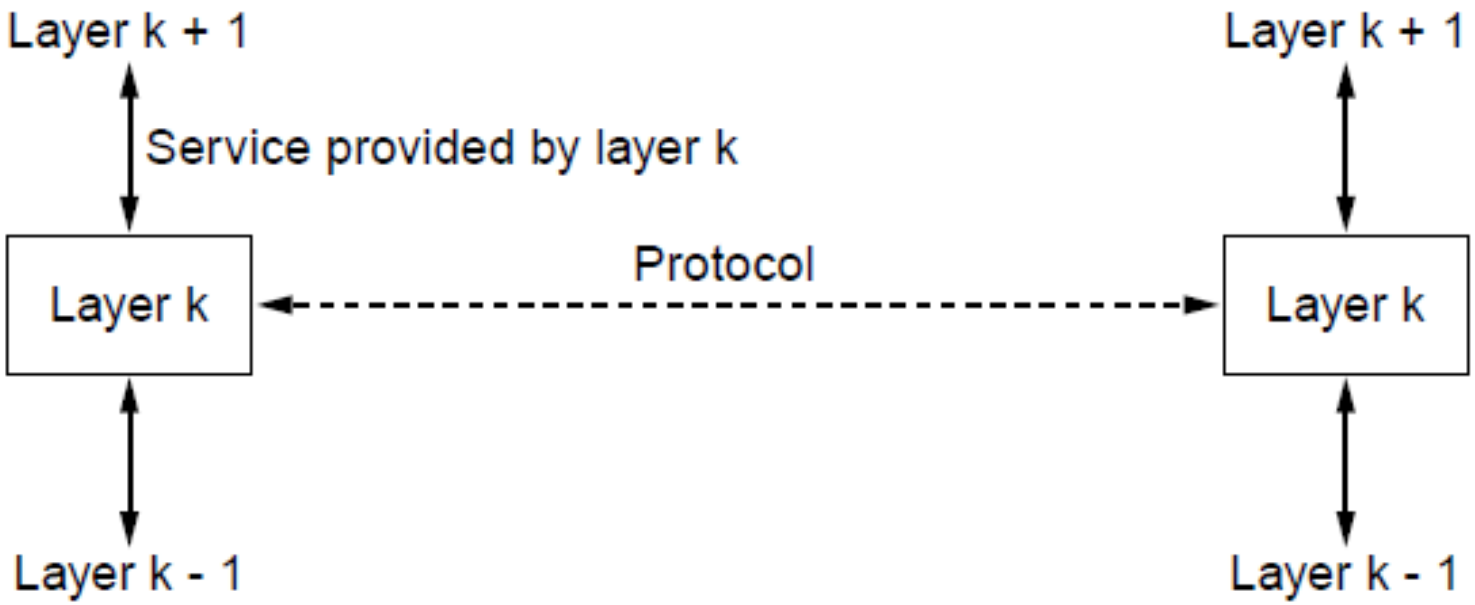
The Relationship of Services to Protocols: A service is a set of primitives provided by a layer to the layer above it. A protocol is a set of rules governing the format and meaning of packets exchanged by peer entities within a layer.

Service Primitives & Protocols

A simple client-server interaction using acknowledged datagrams



The Relationship of Services to Protocols



Reference Models

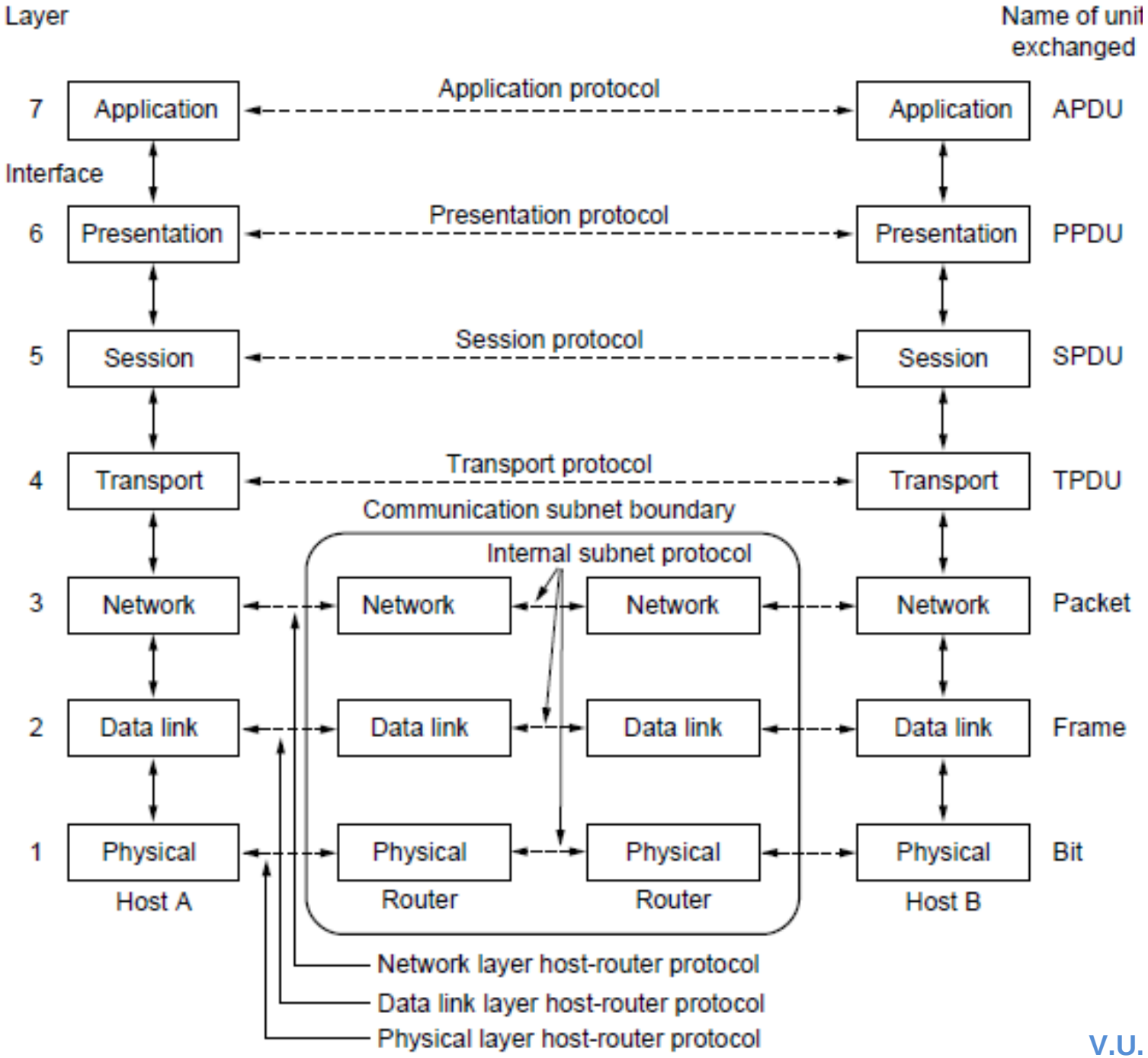
OSI vs TCP/IP: Architectures, Comparisons, and Critiques.

The OSI Reference Model

Principles: Layers created for different abstractions, perform well-defined functions (based on international standard protocols), minimize information flow across boundaries, optimum number of layers.

Layer	Name	Function Focus
7	**Application**	Network services for user applications.
6	**Presentation**	Data representation and formatting.
5	**Session**	Interhost communication and dialog control.
4	**Transport**	End-to-end connections and reliability.
3	**Network**	Logical addressing and routing.
2	**Data link**	MAC addressing and reliable frames.
1	**Physical**	Raw bit stream over physical media.

The OSI Reference Model



The TCP/IP Reference Model

The TCP/IP Reference Model : A practical 4-layer model with core protocols (IP, TCP, UDP).

Layer	Name	Example Protocols
4	**Application**	HTTP, SMTP, RTP, DNS
3	**Transport**	TCP, UDP
2	**Internet**	IP, ICMP
1	**Link**	DSL, SONET, 802.11, Ethernet

The Model Followed in this Course: A hybrid 5-layer model (Application, Transport, Network, Data Link, Physical) is used for study.

Comparison & Critique of Models

Comparison of OSI and TCP/IP

Concepts central to the OSI model are explicitly separated:

- Services: What the layer does.
- Interfaces: How to access it.
- Protocols: How it is implemented.

TCP/IP did not originally clearly distinguish these concepts, making it less suitable for teaching, but its protocols are widely used.

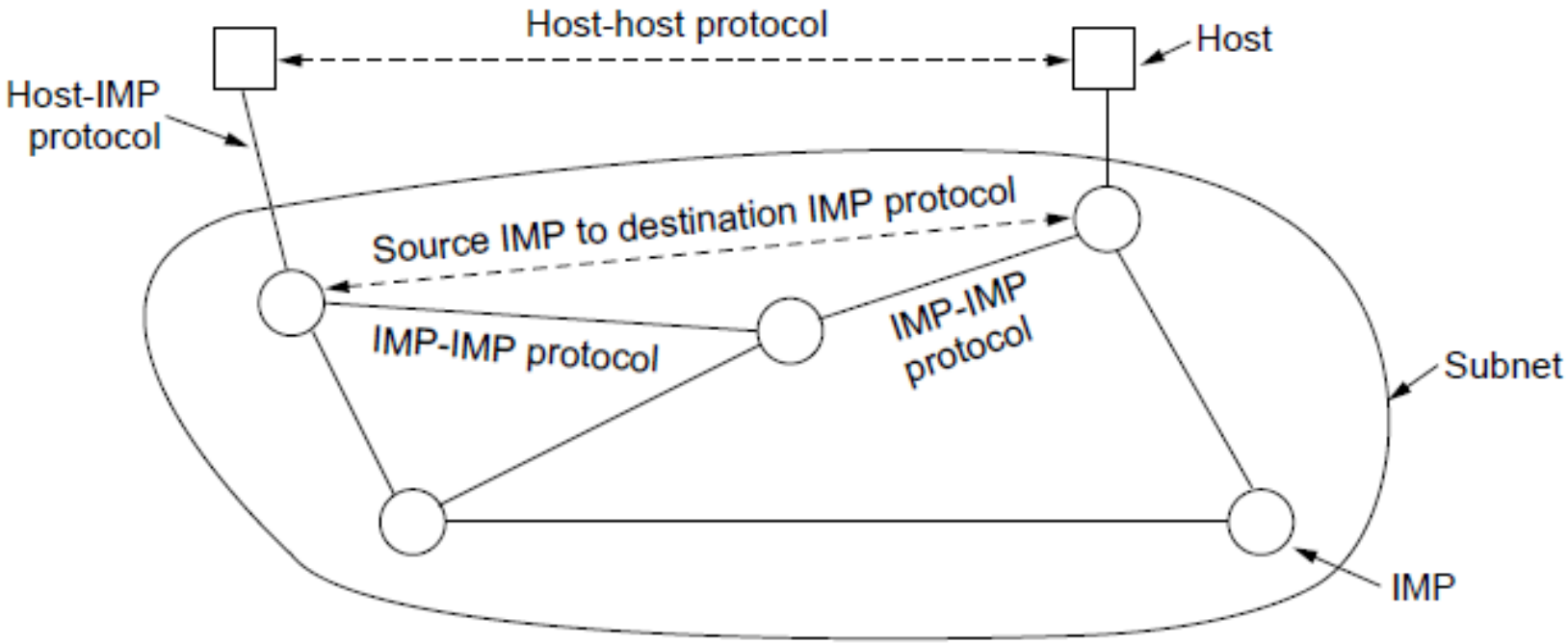
Critique of the OSI Model and Protocols

- Bad timing: "The apocalypse of the two elephants" - standard arrived too late.
- Bad technology: Overly complex, empty layers (Session, Presentation).
- Bad implementations: Huge, unwieldy codebases initially.
- Bad politics: Seen as government/telecom mandates vs researcher-driven TCP/IP.

Example Networks

From the ARPANET to modern wireless, cellular, and sensor networks.

Evolution of The ARPANET



ARPANET Original Design

Evolution of The ARPANET

The ARPANET (1) & (2): The original ARPANET design connected university hosts via IMPs (Interface Message Processors).

Dec 1969

ARPANET (3): Initial 4-node network goes live connecting UCLA, UCSB, SRI, and Utah.

July 1970

ARPANET (3): Network growth expands to East Coast nodes like BBN and MIT.

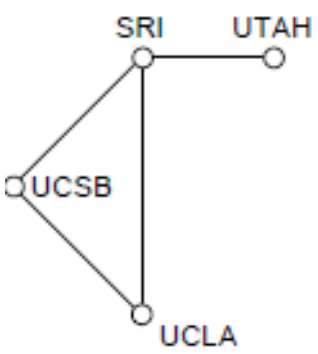
March 1971

ARPANET (3): Significant expansion mapping across the US.

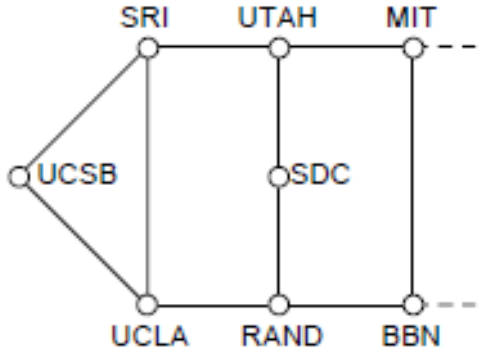
1972

ARPANET (4): April & Sept 1972 show massive nationwide growth, leading to modern Internet.

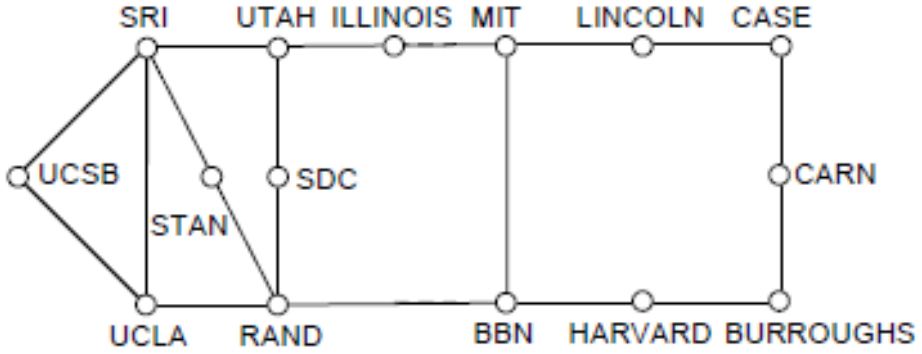
Evolution of The ARPANET



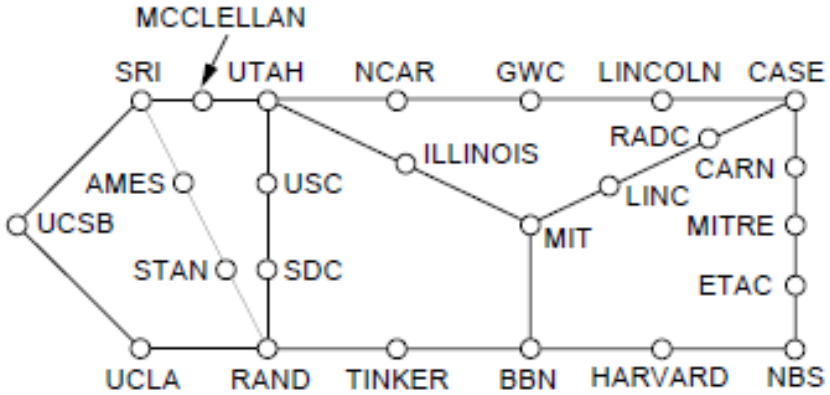
(a) Dec 1969



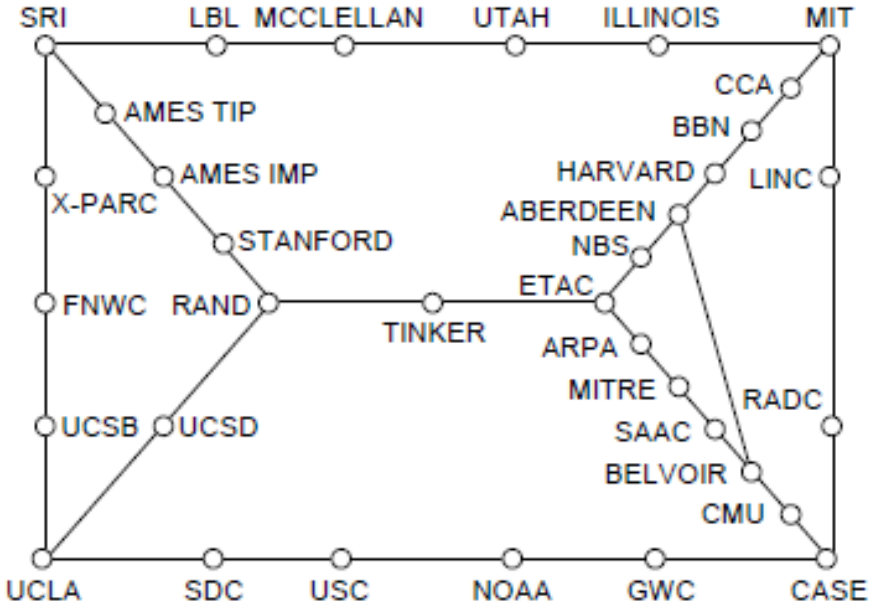
(b) Jul 1970



(c) Mar 1971



(d) Apr 1972



(e) Sept 1972

NSFNET & Internet Architecture



NSFNET

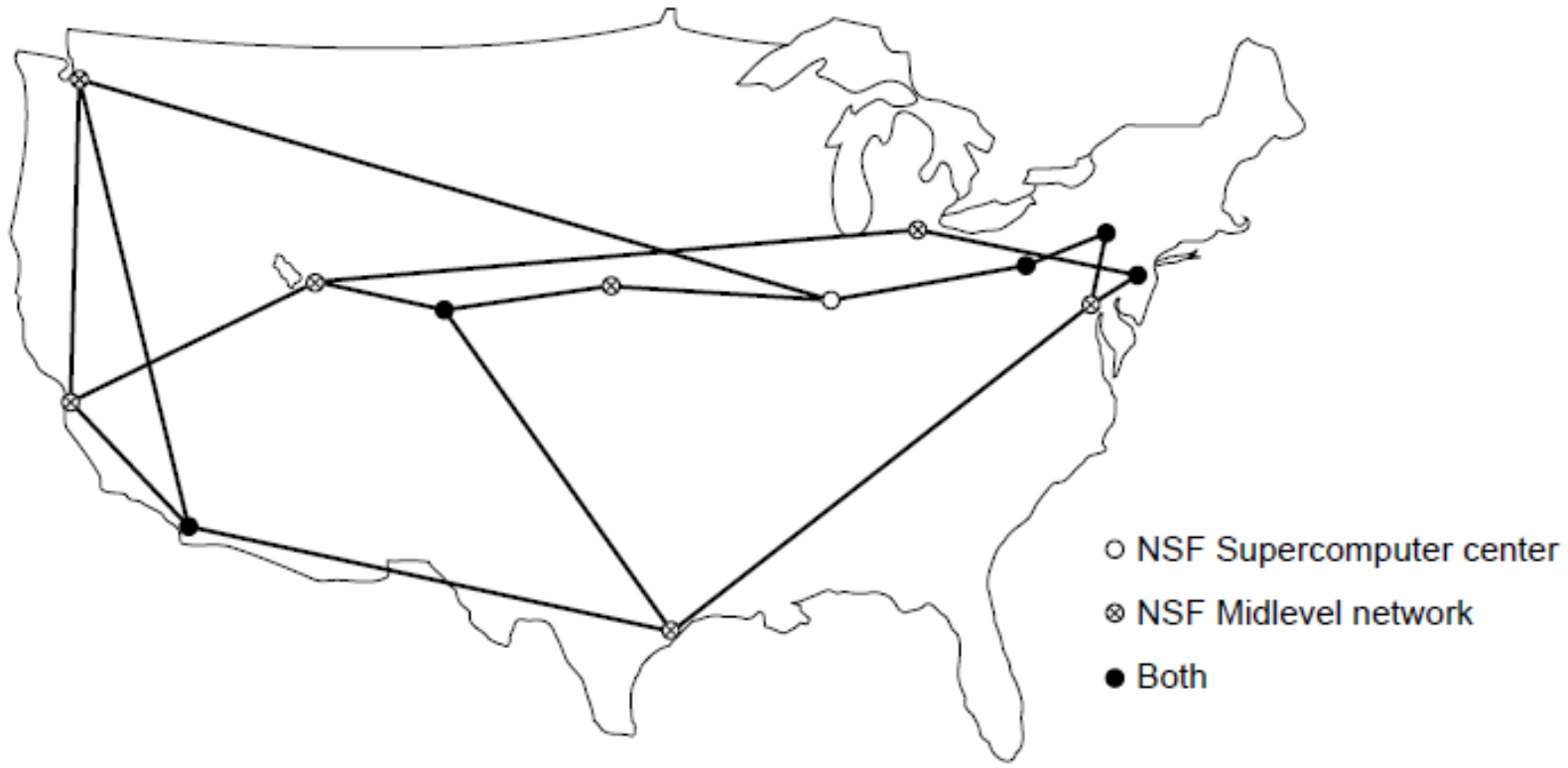
The NSFNET backbone in 1988 replaced ARPANET as the primary US academic network, providing high-speed connectivity to regional networks.



Architecture of the Internet

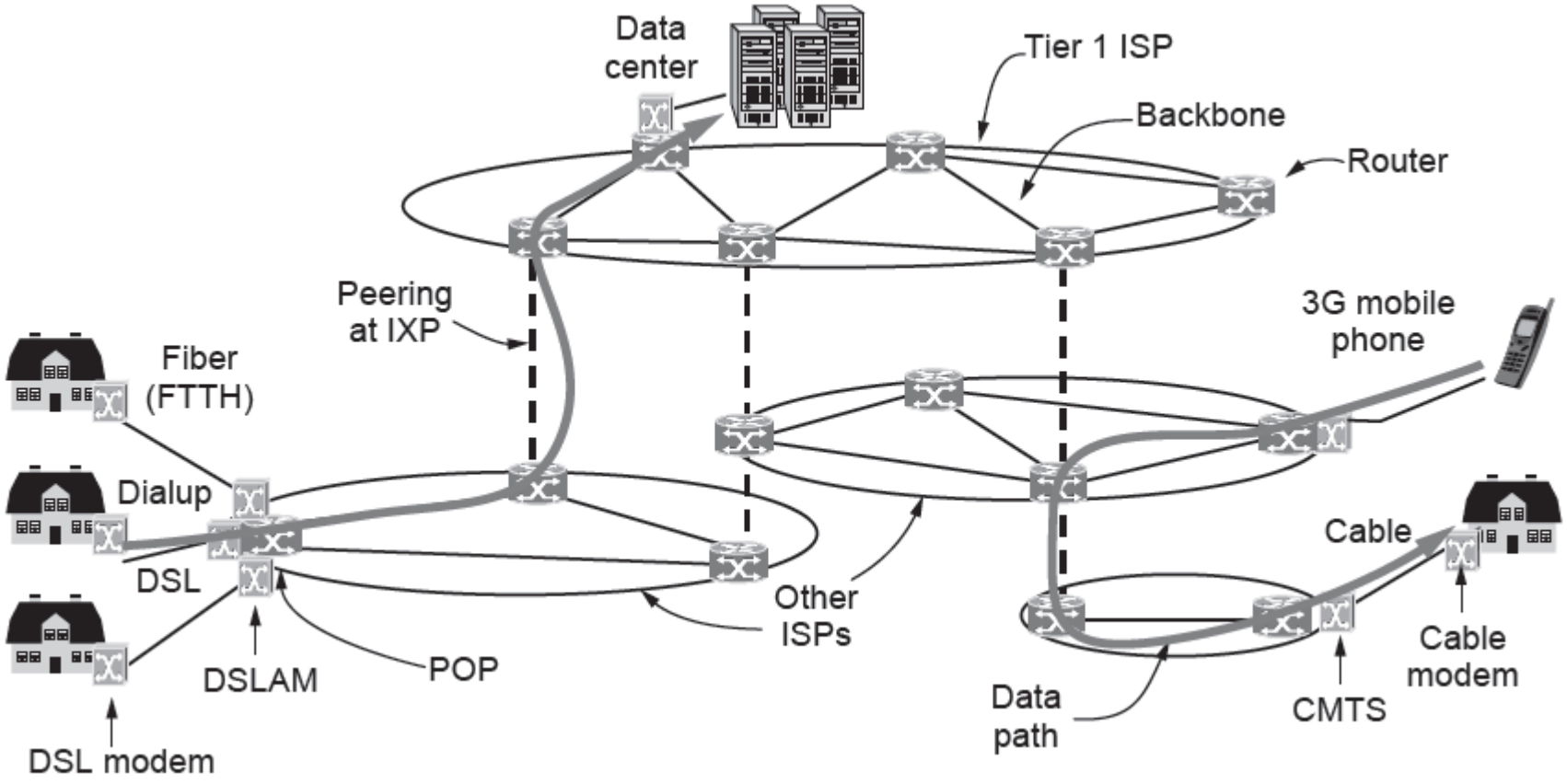
Overview of the Internet architecture: A massive collection of interconnected networks (ISPs, regional, enterprise) held together by the TCP/IP protocol suite.

NSFNET

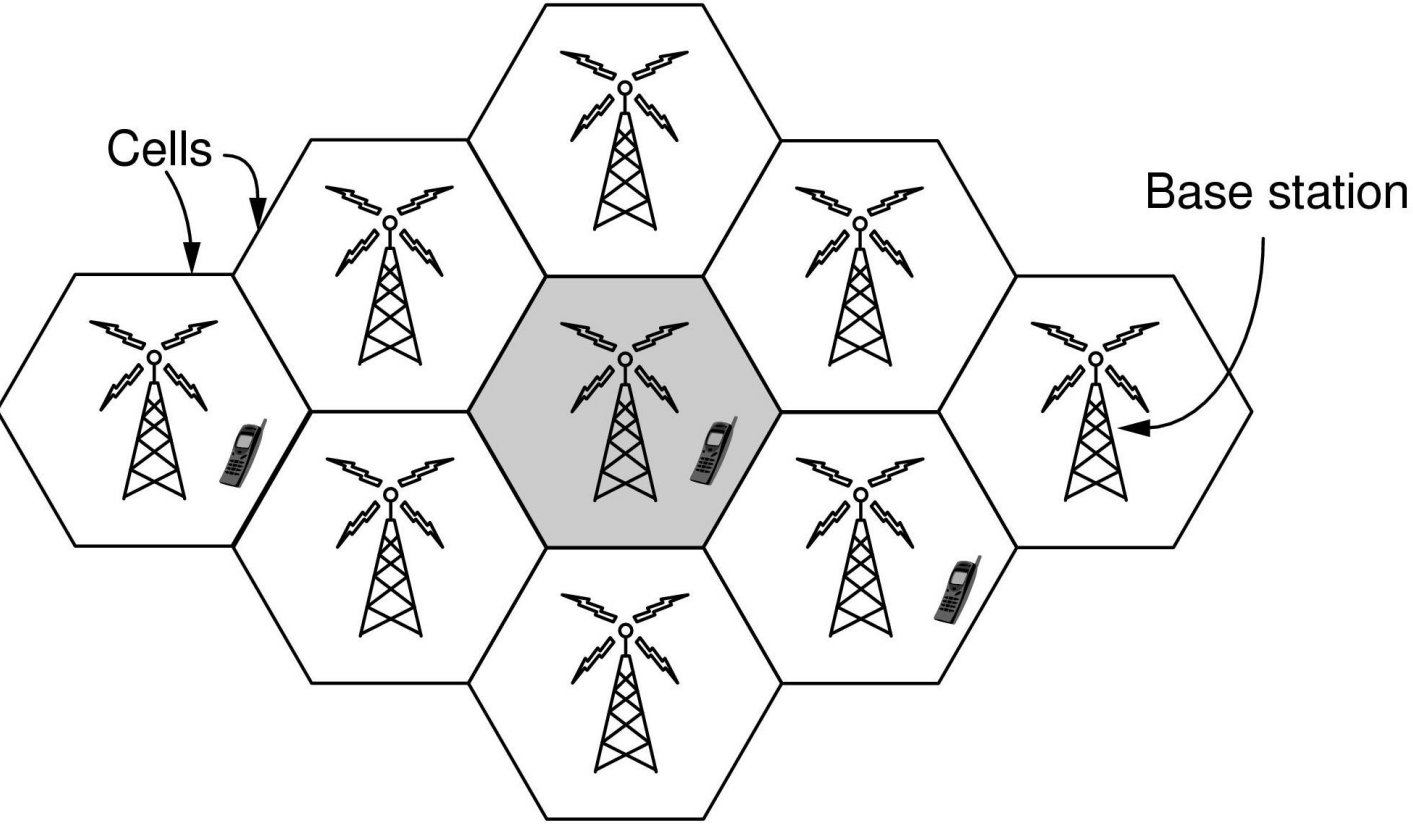


1988

Architecture of the Internet



Third-Generation Mobile Phone Networks



3G Mobile Phone Networks (1): Relies on the cellular design of mobile phone networks using geographic "cells".

3G Mobile Phone Networks (2): The architecture of the UMTS 3G mobile phone network integrates voice and high-speed data.

3G Mobile Phone Networks (3): Mobile phone handover requires seamless transition of connections from one cell tower to another (a) before, and (b) after crossing boundaries.

Wireless LANs: 802.11



Network Types (1)

Wireless network with an access point (infrastructure mode) vs. an Ad hoc network (peer-to-peer without central AP).



Fading (2)

Multipath fading is a major challenge, where signals bounce off objects and arrive at the receiver at different times, causing interference.



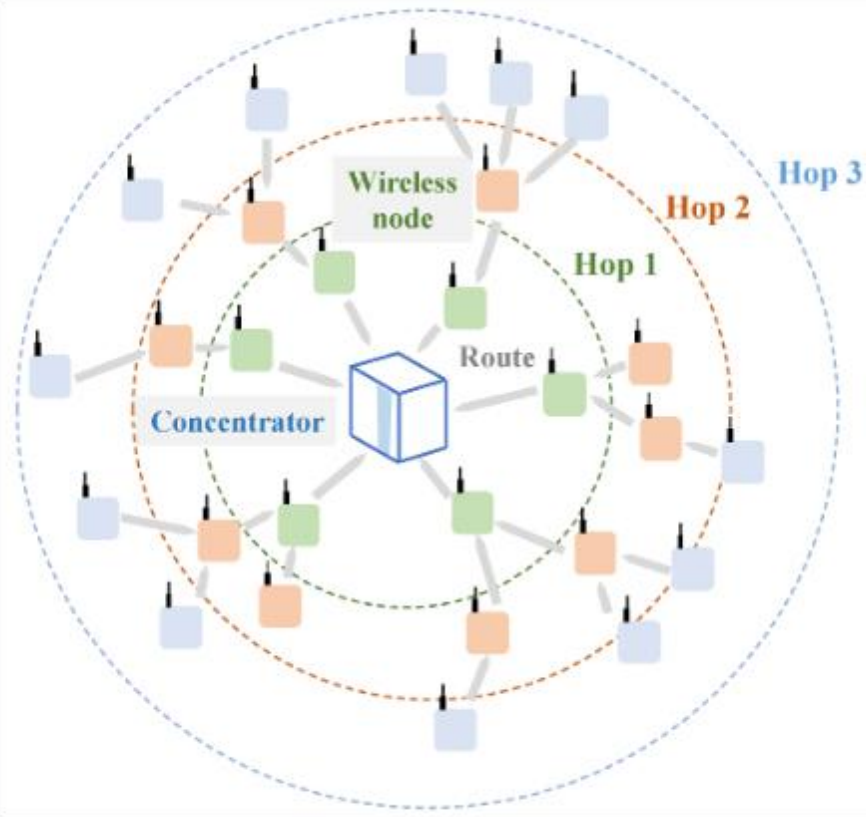
Radio Range (3)

The range of a single radio may not cover the entire system, requiring multi-cell configurations or repeaters.

RFID and Sensor Networks

RFID and Sensor Networks (1): RFID (Radio Frequency Identification) is used to network everyday objects, reading passive tags without batteries.

RFID and Sensor Networks (2): Sensor networks utilize a multihop topology where nodes cooperatively pass data from sensors back to a base station.



Standardization & Metric Units

Network Standardization: Who's Who in International Standards (ITU, IEEE, IETF) ensure equipment compatibility globally.

Metric Units (1) & (2): The principal metric prefixes used in networking (Kilo, Mega, Giga, Tera, etc.). Remember: Telecomm uses powers of 10 (1 Kbps = 1000 bps), while RAM uses powers of 2.

Prefix	Value (Networking/Telecom)
Kilo (K)	10^3 (1,000)
Mega (M)	10^6 (1,000,000)
Giga (G)	10^9 (1,000,000,000)
Tera (T)	10^12 (1,000,000,000,000)

Standardization

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs (WiFi)
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number; nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth, Zigbee)
802.16 *	Broadband wireless (WiMAX)
802.17	Resilient packet ring
802.18	Technical advisory group on radio regulatory issues
802.19	Technical advisory group on coexistence of all these standards
802.20	Mobile broadband wireless (similar to 802.16e)
802.21	Media independent handoff (for roaming over technologies)
802.22	Wireless regional area network

Questions?

End of Chapter 1
